

Synopsis

Prediction of Congestion in Campus Wi-Fi Networks using Machine Learning

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- **Problem Description:**

With the increasing dependence on campus Wi-Fi networks for academic, administrative, and personal use, network congestion has become a critical issue affecting performance, user experience, and service reliability. Congestion occurs when network demand exceeds available capacity, leading to high latency, packet loss, and reduced bandwidth efficiency. Identifying and predicting such congestion in advance is essential for effective network management.

This project aims to design and develop an end-to-end machine learning classification system that analyzes network performance data—such as number of connected devices, latency, packet loss rate, bandwidth usage, and access point load—to predict congestion levels. The system involves data preprocessing, exploratory data analysis to understand patterns and anomalies, feature selection for improving model accuracy, and building predictive models.

In real life, this solution helps network administrators proactively manage traffic, optimize resource allocation, and ensure smooth connectivity across campus environments. It supports better decision-making, reduces downtime, and enhances overall user satisfaction by maintaining efficient and reliable Wi-Fi services.

- **Steps of Implementation:**

This project follows a structured machine learning pipeline for campus Wi-Fi congestion prediction. Each phase of implementation is explained below:

1. User Authentication

A basic interface or system access control is implemented to ensure only authorized users (e.g., network administrators) can access the Wi-Fi congestion prediction system. This maintains data security and controlled usage.

2. Record Handling

The Wi-Fi dataset is collected and managed efficiently. Records containing features like connected devices, latency, packet loss, bandwidth usage, and access point load are stored, updated, and organized for further processing.

3. String Manipulation

Any textual or categorical data (if present, such as access point IDs or labels) is cleaned and formatted. Labels like “Congested” and “Non-Congested” are standardized and converted into machine-readable format.

4. File Handling

Dataset files (CSV/Excel) are read, written, and managed using file handling techniques. Cleaned and processed data is stored for reuse, and model outputs/results are saved for reporting.

5. Main Driver Module

This is the core part of the system where:

- Data preprocessing is performed
- EDA and correlation analysis are conducted
- Feature selection techniques (PCA, Forward, Backward) are applied
- Machine learning models are trained and tested
- Predictions for Wi-Fi congestion are generated

6. Report & Presentation

The final results are presented using performance metrics (accuracy, etc.) and visualizations. Insights such as congestion patterns and model effectiveness are summarized to help in better network management and decision-making.

- **Proposed Modules:**

- 1. Data Collection Module**

Handles the selection and loading of the campus Wi-Fi dataset containing features like connected devices, latency, packet loss, bandwidth usage, and access point load.

- 2. Data Preprocessing Module**

Cleans the dataset by handling missing values, removing duplicates, and applying normalization/standardization to ensure consistency and quality of data.

- 3. Exploratory Data Analysis (EDA) Module**

Performs statistical analysis and visualizations (histograms, scatter plots, box plots, heatmaps) to understand data patterns, detect anomalies, and identify congestion trends.

- 4. Feature Selection Module**

Selects the most relevant features using techniques like PCA, Forward Selection, and Backward Selection to improve model accuracy and reduce redundancy.

- 5. Model Training Module**

Implements machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, KNN, and SVM to train models for congestion prediction.

- 6. Model Evaluation Module**

Evaluates model performance using metrics like accuracy, precision, recall, F1-score, and confusion matrix to select the best model.

- 7. Prediction Module**

Uses the trained model to predict Wi-Fi congestion based on new or real-time network data inputs.

- 8. Reporting & Visualization Module**

Generates reports and visual outputs to present insights, model results, and congestion patterns for better decision-making by network administrators.

- **Required topics from the subject:**

- 1. File Handling**

File handling is used to read and write the Wi-Fi dataset stored in CSV/Excel format. It helps in loading raw data, saving cleaned data, and storing intermediate results during preprocessing and EDA.

- 2. Data Structures (Arrays/DataFrames)**

Data structures like arrays and DataFrames (in Python) are used to store and manipulate large amounts of network data efficiently. They allow easy access, modification, and analysis of features such as latency and bandwidth.

- 3. Data Preprocessing Techniques**

Includes handling missing values, removing duplicates, and scaling data using normalization or standardization. These techniques improve data quality and ensure better performance of analysis.

- 4. Exploratory Data Analysis (EDA)**

EDA involves analyzing data using statistical summaries and visualizations like histograms, scatter plots, and box plots. It helps in understanding patterns, detecting anomalies, and identifying congestion trends.

- 5. Basic Statistics**

Statistical concepts like mean, median, standard deviation, and correlation are used to summarize and understand the dataset. They help in identifying relationships between network features and congestion.

- 6. Visualization Techniques**

Graphs and plots are used to represent data visually for better understanding. Tools like matplotlib and seaborn help in creating heatmaps, histograms, and scatter plots during EDA.

- **Platform Required:**

Jupyter Notebook is used as the primary platform for implementing the project. It provides an interactive environment for writing and executing Python code, performing data preprocessing, and conducting Exploratory Data Analysis (EDA). It also supports visualization libraries like matplotlib and seaborn, making it suitable for analyzing and presenting Wi-Fi network data effectively.

- **Books and Link Sources:**

- 1. Pandas Documentation (Data Handling & Preprocessing)**

<https://pandas.pydata.org/docs/>

Used for reading datasets, handling missing values, and performing data preprocessing.

- 2. Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow**

By - Aurélien Géron (Book)

A comprehensive book covering the complete ML pipeline including preprocessing, EDA, feature selection, model building, and evaluation. Useful for understanding end-to-end system design.

- 3. Scikit-learn Official Documentation (ML Models & Preprocessing)**

<https://scikit-learn.org/stable/>

Covers data preprocessing, feature selection methods (PCA), and classification algorithms like Logistic Regression, SVM, KNN, etc.

- 4. Kaggle Learn (Practical ML & EDA Tutorials)**

<https://www.kaggle.com/learn>

Offers hands-on tutorials on data preprocessing, EDA, feature engineering, and machine learning workflows.

- 5. Campus Wi-Fi Network Research Paper (IEEE Xplore / Google Scholar)**

<https://scholar.google.com/>

Used to study real-world applications of network congestion prediction and understand how ML is applied in network analysis. .